Complete inputs for the enlarged Mixes topic

Human Compression project — F005 and L016

This input packet adds Leemans–Toledo's *Faithful and thin non-polytopal maniplexes*, arXiv:2305.01372v2 (10 May 2023), to the existing Mixes topic. All 12 original pages are included. The entire earlier common paper (5 pages), new cover common paper (4 pages), and all three application papers (8, 4 and 6 pages) are charged. There is no overlap deduction or free prior theory.

Primary rule, fixed before counting: NFC characters without whitespace in literal, untrimmed PDF extraction on both sides. The raw column retains headers, footers and the source banner. No factor is evaluated in this packet.

Complete document	Pages	Raw	Selected	Calibrated
F005: complete source	21	57,753	57,720	57,720
L016: complete source	12	35,506	35,127	35,127
Earlier common theory	5	12,615	12,610	12,610
Cover common theory	4	11,641	11,637	11,637
F005 local application	8	20,788	20,780	20,780
F005 exceptional application	4	10,289	10,285	10,285
L016 application	6	15,125	15,119	15,119
Both required certificates	2	3,431	3,429	3,429
New optional validations	3	7,285	7,282	7,282
Old mixed appendix (inventory)	4	8,813	8,809	8,809
Old optional appendix (inventory)	3	7,212	7,209	7,209

The scientific comparison has two complete sources, two complete common papers and three complete application papers. The last four table rows are program representations or historical inventories, not four additive charges. The new two-page required appendix contains both whole certificates. It is an alternative to the two required raw files, never an additional cost. Any later conditional application comparison must declare both whole common papers and both required certificates already acquired.

Optional verification is symmetric. The four optional validation programs may enter a comparison only when corresponding source-side optional verification is also included; otherwise optional verification is excluded on both sides. Historical software is unavailable, so its cost is unknown, not zero. Neither source prints an executable listing; its written algorithms and formulas stay in the full text. The earlier mixed appendices and earlier optional-inclusive numerical results remain unchanged historical inventories. They are not relabelled as current required-only compression factors.

Input reuse and scientific scope. All 180 PDF cells and 70 raw-file cells from the accepted F005 input audit are retained exactly, without reextracting those documents. The new extraction adds 150 PDF cells over 27 pages and 60 raw-file cells. The 35 F005 and 19 L016 dispositions describe two originals, not 54 sources. L016's literal thin-base Lemma 3.5 is disproved and repaired under the stated faithful-base hypothesis. Its principal family and rank-three results are recovered; the small-example census, atlas claims and further-cover questions remain visibly outside the contract. All three final F005 problems remain unanswered. Distinct scientific reviews and root adoptions are bound; distinct measurement-input acceptance is still required.

Whole files and representation limits

Complete raw file	Role	NFC nonspace
F005: `check_coprime.g`	Required certificate	1,871
F005: `check_single_base.py`	Optional validation	6,127
F005: `check_exceptional.py`	Optional validation	6,610
F005: `check.py`	Driver	7,238
F005: `check_integrity.py`	Driver	1,827
F005: `check.py`	Additive driver	4,609
F005: `FiberIntersections.lean`	Partial Lean	1,714
L016: `base_certificate.g`	Required certificate	1,036
L016: `validate_base.g`	Optional validation	1,847
L016: `validate_covers.py`	Optional validation	4,883
L016: `check.py`	Driver	7,055
L016: `check_integrity.py`	Driver	1,270
L016: `CoverFibers.lean`	Partial Lean	2,467

The two required GAP files identify the finite coprime-mix example and the order-96 marked presentation. The four optional programs check written coordinate constructions. Five operational files and two Lean files remain separate inventories. The Lean scope is six generic declarations (four new, two reused), not a formalization of the complete source or applications. Complete-file inclusion in both new appendices is checked without line pruning. A comparison including required programs must disclose the unknown source-side software cost; it cannot claim a complete measured software comparison.

Units and extraction. Every PDF has five units under NFC and NFKC in three variants: literal untrimmed, literal selected, and calibrated selected. The five units are whitespace tokens, alphanumeric units, nonspace characters, characters with collapsed whitespace, and UTF-8 bytes with collapsed whitespace. Selected L016 extraction removes only its isolated first-page footer and later running headers; replacement selection removes isolated page footers. The source banner stays. Raw files use all five units and both normalizations.

The source has 23 actual Type1 dictionaries and 566 encoded glyph classes. Fifteen replacement Type0 dictionaries retain literal extraction. Actual embedded outlines were inspected for 19 exceptional classes. Twelve observed aliases occur 331 times: script letters, brace components and four arrow-tee components. Only the calibrated sensitivity changes those representations; no guessed composite arrow is inserted. UTF-8 differences do not measure semantic proof complexity. The pinned extractor uses `pypdf 6.13.0` and `fontTools 4.63.0`, checks ordinary returned text, and separates actual glyph events from generated line breaks. No Form callback duplication was found.

Visual and external limits. All 27 new source, proof and appendix pages were viewed; the earlier 45-page record is reused with its accepted bindings. The four source figures on pages 7, 8, 9 and 11 retain their extractable labels and captions. Edge strokes, colors, circles and layout have no established full cost. Historical software, retained external proofs, runtime and complete formal verification likewise remain unknown. No source text, font bytes or page images are exported. The portable replay checks the inputs, 47 rejection controls and fresh reader/appendix renders; it does not rerun the scientific GAP or Lean work. The measurement author also authored L016, so input acceptance must be distinct.